

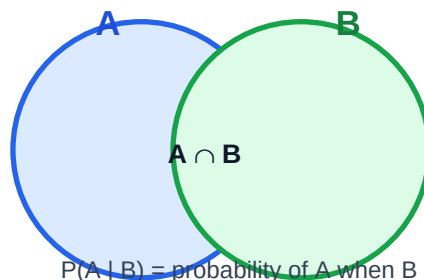
■ Part A: Probability & Conceptual Foundation

Exam-ready answers • Clear theory • Visual diagrams • 10/10 presentation

1. What is Conditional Probability?

Conditional Probability is the probability of an event occurring when another event is already known to have occurred. It helps us update the probability of one event using available information about another event.

$$P(A | B) = P(A \cap B) / P(B), \text{ where } P(B) > 0$$



2. Explain Bayes' Theorem and its importance in classification problems.

Bayes' Theorem calculates the probability of a hypothesis after observing evidence. In classification, it combines prior knowledge with new features to estimate the probability of each class and choose the most likely class.

$$P(A | B) = [P(B | A) \times P(A)] / P(B)$$



Update belief using observed evidence

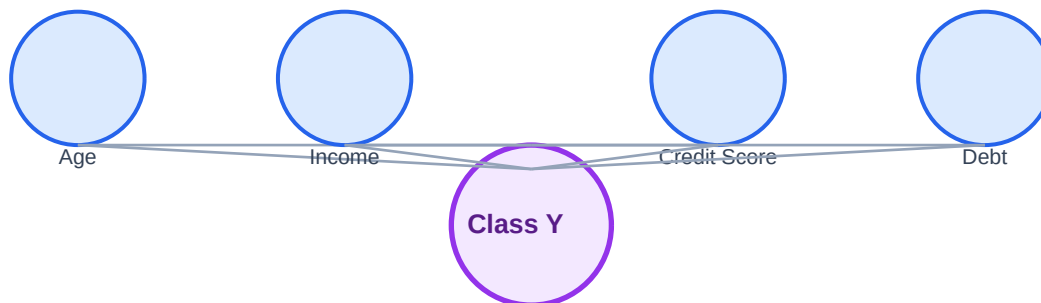
$$\text{Posterior} = (\text{Likelihood} \times \text{Prior}) / \text{Evidence}$$

Why important? It is the mathematical foundation of **Naive Bayes** and is useful when classification decisions must be based on probabilities.

3. What assumptions does the Naive Bayes Classifier make?

The main assumption is that the input features are **conditionally independent** given the class. In other words, once the class is known, one feature does not affect another feature's probability.

Example: For credit-risk prediction, Age, Income, and Credit Score are treated as independent when the risk class is fixed.



Features are treated as conditionally independent given the class

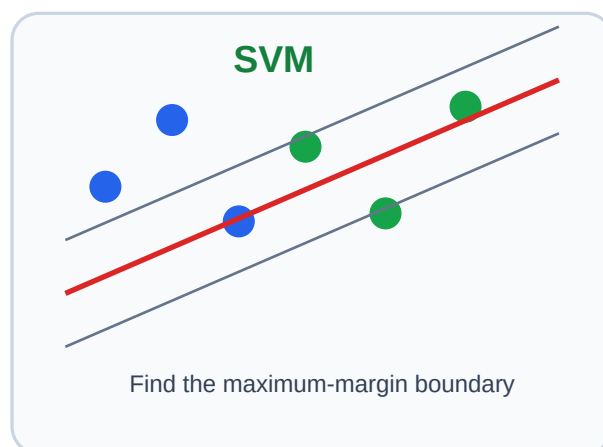
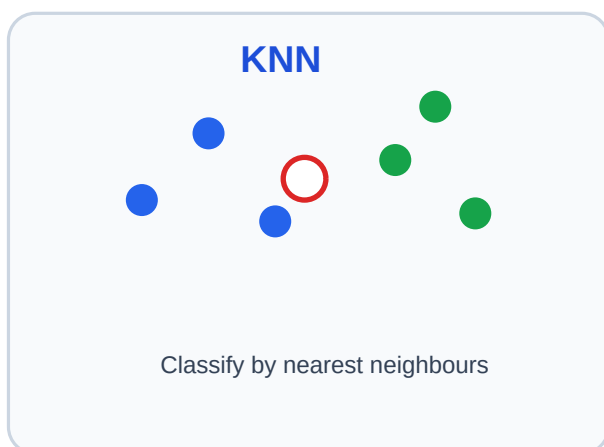
4. Explain the working principle of K-Nearest Neighbors (KNN) and Support Vector Machine (SVM).

■ K-Nearest Neighbors (KNN)

KNN is a **distance-based** algorithm. It finds the **K nearest training points** to a new observation using a distance measure such as Euclidean or Manhattan distance, then assigns the class based on majority voting.

■ Support Vector Machine (SVM)

SVM is a **margin-based** algorithm. It finds the best decision boundary (hyperplane) that separates classes while maximizing the margin between the boundary and the closest observations called **support vectors**.



5. Compare distance-based, probabilistic, and margin-based classifiers.

These classifiers use different decision principles: **distance-based** models classify using similarity, **probabilistic** models classify using estimated probabilities, and **margin-based** models classify using an optimal decision boundary.

Distance-based

Uses similarity / distance

KNN

Probabilistic

Uses probability

Naive Bayes

Margin-based

Uses decision boundary

SVM

Type	Core idea	Example	Decision basis
Distance-based	Nearby points are more similar	KNN	Distance / neighbours
Probabilistic	Estimate probability of each class	Naive Bayes	Posterior probability
Margin-based	Separate classes with maximum margin	SVM	Decision boundary / margin

■ Quick Revision

Conditional Probability → probability with known information

Bayes' Theorem → updates probability using evidence

Naive Bayes → assumes conditional independence

KNN → nearest neighbours + voting

SVM → maximum-margin decision boundary

Presentation tip: In an exam, write the definition first, then the key formula/principle, and finish with one short example or application. This structure makes the answer clear, complete, and easy to evaluate.